Lab 7: Fuzzing for Input Validation Bugs

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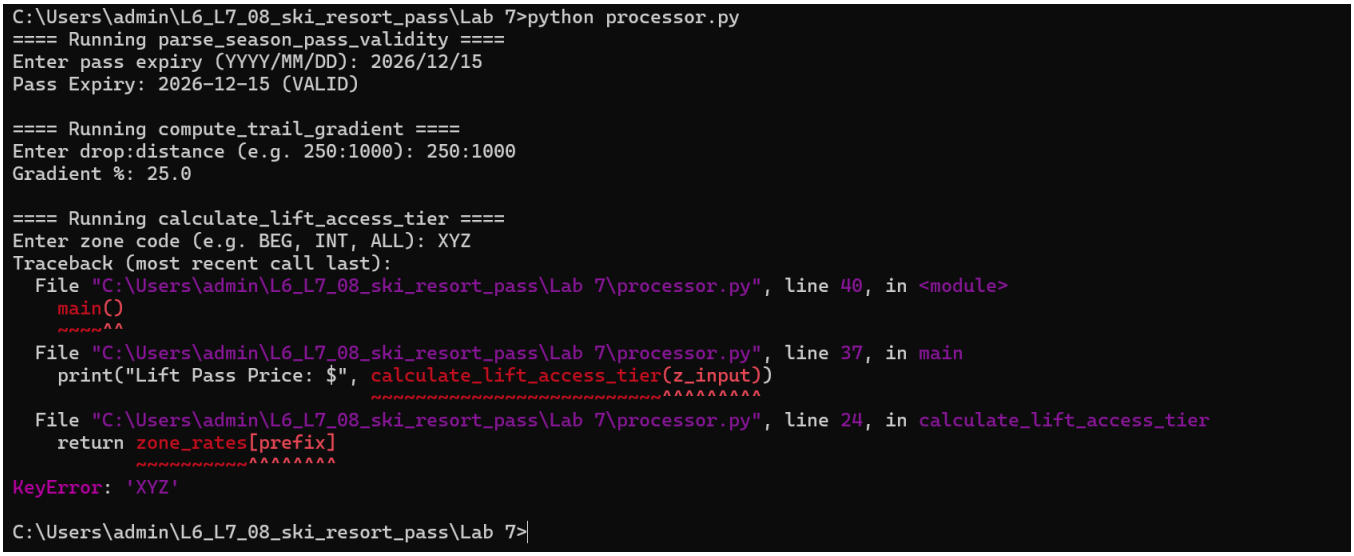
Tools: Hypothesis 6.170.15 + pytest 9.1.1 on Python 3.13.7 (Windows). Module under test: `processor.py` (`parse_season_pass_validity`, `compute_trail_gradient`, `calculate_lift_access_tier`).

Step 1: Install dependencies

`pip install hypothesis` and `pip install pytest`.

Step 2: Run the original (buggy) processor.py

`python processor.py` works for a valid date and gradient, but the unknown zone code **XYZ** crashes the program with an unhandled `KeyError`.



Step 3: Complete test_processor.py

The three required tests wrap each function call in try/except using `st.text() | st.none()` and fail the test if any exception escapes. Eight extra property tests check correct output for valid inputs, return types, and zero distance.

Step 4: Run test_processor.py against the buggy code

`python -m pytest test_processor.py` result: **7 failed, 4 passed**. Start of the output:

```
C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7>python -m pytest test_processor.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.7, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7
plugins: anyio-4.16.0, hypothesis-6.170.15, langsmith-0.14.17, locust-2.46.7, asyncio-0.24.0, cov-6.2.1
asyncio: mode=Mode.STRICT, default_loop_scope=None
collected 11 items

test_processor.py FFF..F.FF.F                                                                                                        [100%]

================================================================= FAILURES ==================================================================
_____________________________________________ test_parse_season_pass_validity_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 9, in test_parse_season_pass_validity_no_crash
  |     def test_parse_season_pass_validity_no_crash(s):
  |                ^^^
  |   File "C:\Users\admin\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 11, in test_parse_season_pass_validity_no_crash
    |     parse_season_pass_validity(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 4, in parse_season_pass_validity
    |     month = int(parts[1])
    |             ~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 13, in test_parse_season_pass_validity_no_crash
    |     assert False, f"parse_season_pass_validity crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_season_pass_validity crashed with: IndexError('list index out of range') for input '0'
    | assert False
    | Failing test case: test_parse_season_pass_validity_no_crash(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 11, in test_parse_season_pass_validity_no_crash
    |     parse_season_pass_validity(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 3, in parse_season_pass_validity
    |     year = int(parts[0])
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 13, in test_parse_season_pass_validity_no_crash
    |     assert False, f"parse_season_pass_validity crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_season_pass_validity crashed with: ValueError("invalid literal for int() with base 10: ''") for input ''
    | assert False
    | Failing test case: test_parse_season_pass_validity_no_crash(
    |     s='',  # or any other generated value
```

Summary of the same run:

```
    |             ~~~~~~~~~~^^^^^^^^
    | KeyError: ''
    | Failing test case: test_tier_returns_none_or_known_price(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 81, in test_tier_returns_none_or_known_price
    |     assert calculate_lift_access_tier(s) in (None, 35.0, 60.0, 85.0, 110.0)
    |             ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 17, in calculate_lift_access_tier
    |     prefix = zone_code.strip().upper()[:3]
    |              ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    | Failing test case: test_tier_returns_none_or_known_price(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
========================================================== short test summary info ==========================================================
FAILED test_processor.py::test_parse_season_pass_validity_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_compute_trail_gradient_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_calculate_lift_access_tier_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_parse_returns_string - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_gradient_returns_none_or_finite_float - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_gradient_zero_distance_is_handled - ZeroDivisionError: float division by zero
FAILED test_processor.py::test_tier_returns_none_or_known_price - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
========================================================= 7 failed, 4 passed in 19.24s ==========================================================

C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7>
```

Bugs found by Hypothesis (minimal failing inputs)

Function	Failing input	Exception	Cause
parse_season_pass_validity	None / '' / '0'	AttributeError / ValueError / IndexError	No type check; assumes 3 numeric parts
parse_season_pass_validity	'2026/02/30', '2026/13/01'	(wrong result)	Month and day only range-checked 1-12 / 1-31; no real calendar validation
compute_trail_gradient	None, '', ':'	AttributeError / ValueError	No type check; assumes exactly 2 float parts
compute_trail_gradient	'0.0:0'	ZeroDivisionError	Distance of zero not handled
compute_trail_gradient	'nan:1', 'inf:1'	(returns nan / inf)	float() accepts nan/inf; result not validated
calculate_lift_access_tier	None	AttributeError	No type check on zone_code
calculate_lift_access_tier	'', 'XYZ'	KeyError	Dictionary lookup with [] on unknown key

Step 5: Fix processor.py and run it

Fixes: (1) `isinstance(..., str)` guards for None / non-string input; (2) exact part-count checks after split; (3) int()/float() wrapped in try/except; (4) dates validated with `datetime.date` (rejects month 13, Feb 30, year 0); (5) gradient rejects NaN/inf, zero or negative distance and overflow, returning `None`; (6) tier lookup uses `dict.get()` and returns `None` for unknown zones; (7) `main()` prints friendly messages instead of crashing. Same inputs as Step 2 now give a clean message instead of a traceback:

```
C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7>python processor.py
==== Running parse_season_pass_validity ====
Enter pass expiry (YYYY/MM/DD): 2026/12/15
Pass Expiry: 2026-12-15 (VALID)

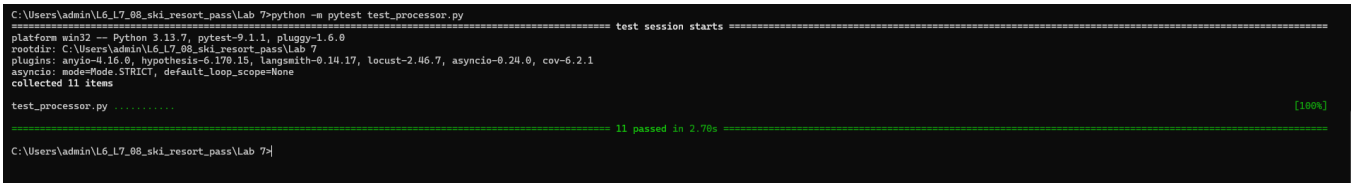
==== Running compute_trail_gradient ====
Enter drop:distance (e.g. 250:1000): 250:1000
Gradient %: 25.0

==== Running calculate_lift_access_tier ====
Enter zone code (e.g. BEG, INT, ALL): XYZ
Lift Pass Price: Invalid zone code (use BEG, INT, ADV or ALL)

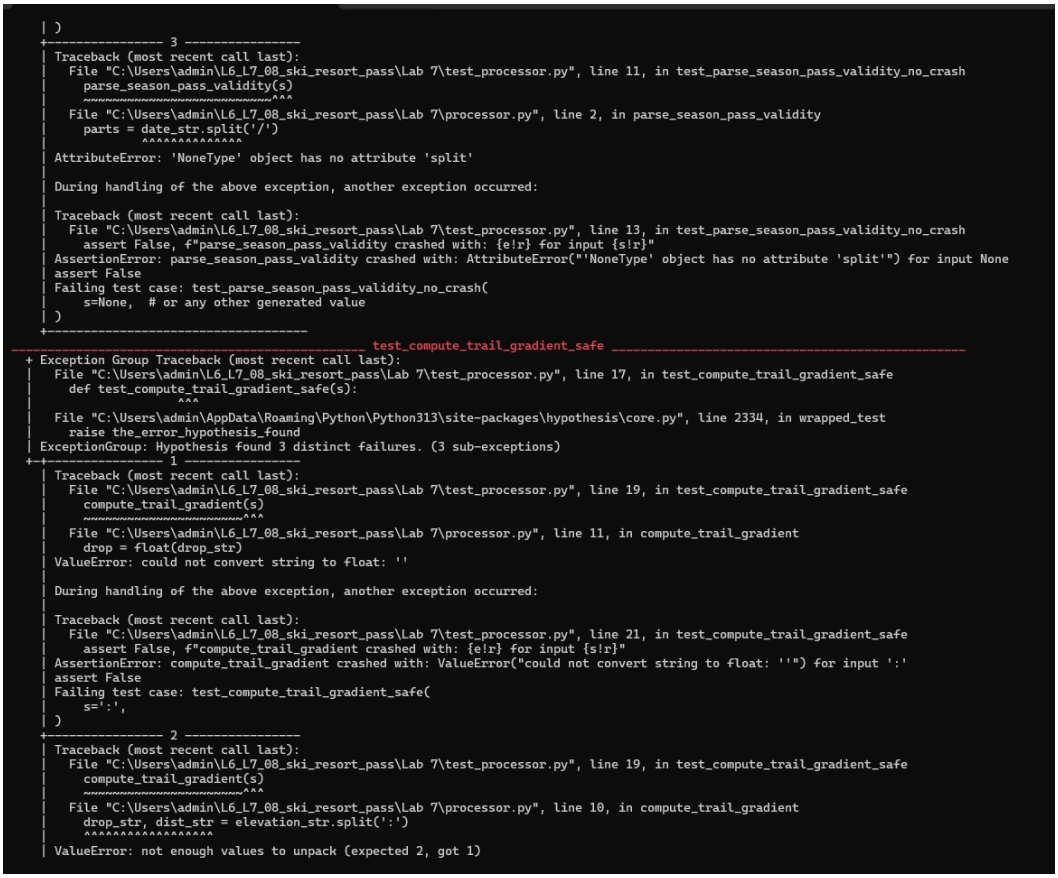
C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7>
```

Step 6: Re-run test_processor.py against the fixed processor.py

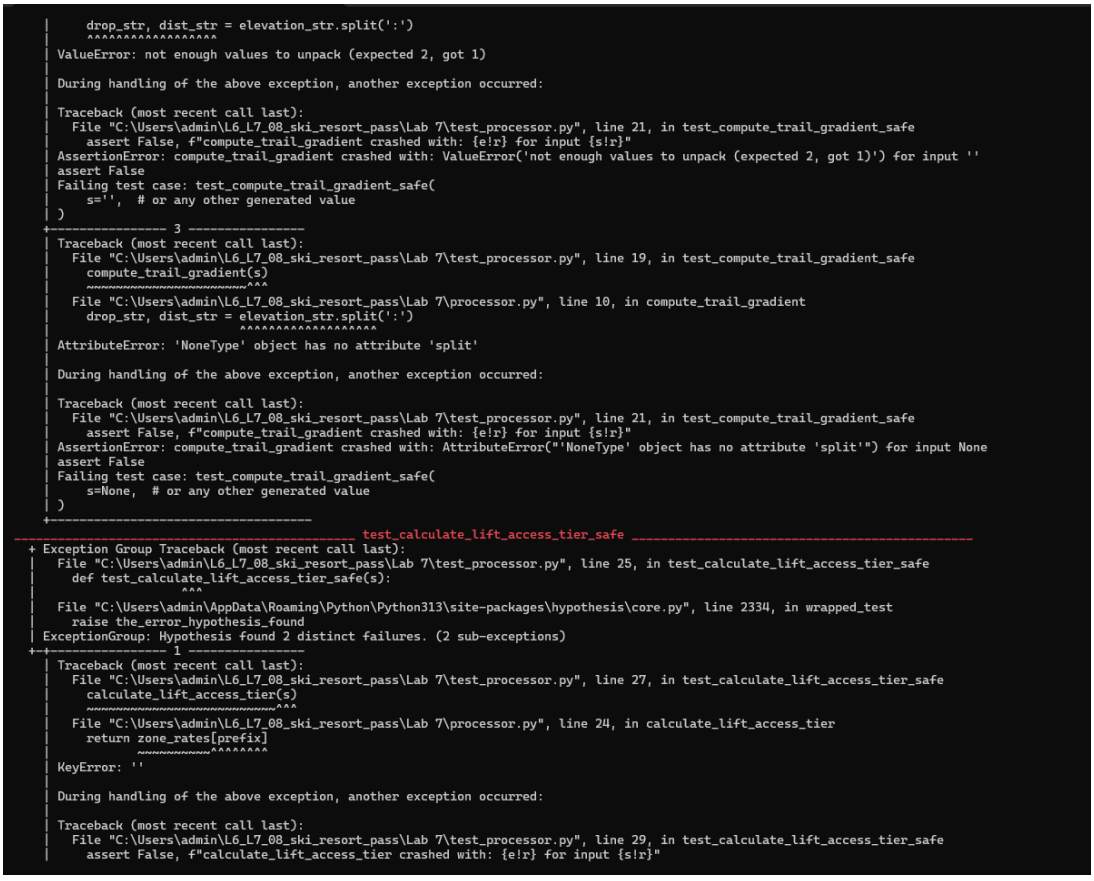
`python -m pytest test_processor.py` result: **11 passed**.



Appendix A: Remaining pytest output from Step 4 (buggy code)



```
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 11, in test_parse_season_pass_validity_no_crash
    |     parse_season_pass_validity(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 2, in parse_season_pass_validity
    |     parts = date_str.split('/')
    |             ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 13, in test_parse_season_pass_validity_no_crash
    |     assert False, f"parse_season_pass_validity crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_season_pass_validity crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
    | assert False
    | Failing test case: test_parse_season_pass_validity_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________ test_compute_trail_gradient_safe _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 17, in test_compute_trail_gradient_safe
  |     def test_compute_trail_gradient_safe(s):
  |                ^^^
  |   File "C:\Users\admin\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 19, in test_compute_trail_gradient_safe
    |     compute_trail_gradient(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 11, in compute_trail_gradient
    |     drop = float(drop_str)
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 21, in test_compute_trail_gradient_safe
    |     assert False, f"compute_trail_gradient crashed with: {e!r} for input {s!r}"
    | AssertionError: compute_trail_gradient crashed with: ValueError("could not convert string to float: ''") for input ':'
    | assert False
    | Failing test case: test_compute_trail_gradient_safe(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 19, in test_compute_trail_gradient_safe
    |     compute_trail_gradient(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 10, in compute_trail_gradient
    |     drop_str, dist_str = elevation_str.split(':')
    |     ^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
```



```
    |     drop_str, dist_str = elevation_str.split(':')
    |     ^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 21, in test_compute_trail_gradient_safe
    |     assert False, f"compute_trail_gradient crashed with: {e!r} for input {s!r}"
    | AssertionError: compute_trail_gradient crashed with: ValueError('not enough values to unpack (expected 2, got 1)') for input ''
    | assert False
    | Failing test case: test_compute_trail_gradient_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 19, in test_compute_trail_gradient_safe
    |     compute_trail_gradient(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 10, in compute_trail_gradient
    |     drop_str, dist_str = elevation_str.split(':')
    |                          ^^^^^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 21, in test_compute_trail_gradient_safe
    |     assert False, f"compute_trail_gradient crashed with: {e!r} for input {s!r}"
    | AssertionError: compute_trail_gradient crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
    | assert False
    | Failing test case: test_compute_trail_gradient_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________ test_calculate_lift_access_tier_safe _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 25, in test_calculate_lift_access_tier_safe
  |     def test_calculate_lift_access_tier_safe(s):
  |                    ^^^
  |   File "C:\Users\admin\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 27, in test_calculate_lift_access_tier_safe
    |     calculate_lift_access_tier(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 24, in calculate_lift_access_tier
    |     return zone_rates[prefix]
    |            ~~~~~~~~~~^^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 29, in test_calculate_lift_access_tier_safe
    |     assert False, f"calculate_lift_access_tier crashed with: {e!r} for input {s!r}"
```

```
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 29, in test_calculate_lift_access_tier_safe
    |     assert False, f"calculate_lift_access_tier crashed with: {e!r} for input {s!r}"
    | AssertionError: calculate_lift_access_tier crashed with: KeyError('') for input ''
    | assert False
    | Failing test case: test_calculate_lift_access_tier_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 27, in test_calculate_lift_access_tier_safe
    |     calculate_lift_access_tier(s)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 17, in calculate_lift_access_tier
    |     prefix = zone_code.strip().upper()[:3]
    |              ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 29, in test_calculate_lift_access_tier_safe
    |     assert False, f"calculate_lift_access_tier crashed with: {e!r} for input {s!r}"
    | AssertionError: calculate_lift_access_tier crashed with: AttributeError("'NoneType' object has no attribute 'strip'") for input None
    | assert False
    | Failing test case: test_calculate_lift_access_tier_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________ test_parse_returns_string ______________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 50, in test_parse_returns_string
  |     def test_parse_returns_string(s):
  |                ^^^
  |   File "C:\Users\admin\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 51, in test_parse_returns_string
    |     assert isinstance(parse_season_pass_validity(s), str)
    |                       ~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 3, in parse_season_pass_validity
    |     year = int(parts[0])
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_parse_returns_string(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 51, in test_parse_returns_string
    |     assert isinstance(parse_season_pass_validity(s), str)
    |                       ~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 2, in parse_season_pass_validity
    |     parts = date_str.split('/')
    |             ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_returns_string(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________ test_gradient_returns_none_or_finite_float ______________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 61, in test_gradient_returns_none_or_finite_float
  |     def test_gradient_returns_none_or_finite_float(s):
  |                ^^^
  |   File "C:\Users\admin\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 62, in test_gradient_returns_none_or_finite_float
    |     result = compute_trail_gradient(s)
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 10, in compute_trail_gradient
    |     drop_str, dist_str = elevation_str.split(':')
    |     ^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_gradient_returns_none_or_finite_float(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 62, in test_gradient_returns_none_or_finite_float
    |     result = compute_trail_gradient(s)
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 10, in compute_trail_gradient
    |     drop_str, dist_str = elevation_str.split(':')
    |     ^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_gradient_returns_none_or_finite_float(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________ test_gradient_zero_distance_is_handled ______________________________________

    @given(st.sampled_from(["0", "0.0", "-0", "0e5", "+0", " 0 "]), st.floats(-1e6, 1e6))
>   def test_gradient_zero_distance_is_handled(dist, drop):
                ^^^

test_processor.py:68:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
test_processor.py:70: in test_gradient_zero_distance_is_handled
    assert compute_trail_gradient(f"{drop}:{dist}") is None
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

elevation_str = '0.0:0'

    def compute_trail_gradient(elevation_str):
```

```
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

elevation_str = '0.0:0'

    def compute_trail_gradient(elevation_str):
        drop_str, dist_str = elevation_str.split(':')
        drop = float(drop_str)
        distance = float(dist_str)
>       gradient_pct = (drop / distance) * 100.0
                        ^^^^^^^^^^^^^^^
E       ZeroDivisionError: float division by zero
E       Failing test case: test_gradient_zero_distance_is_handled(
E           # The test always failed when commented parts were varied together.
E           dist='0',  # or any other generated value
E           drop=0.0,  # or any other generated value
E       )

processor.py:13: ZeroDivisionError
_____________________________________________ test_tier_returns_none_or_known_price _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 80, in test_tier_returns_none_or_known_price
  |     def test_tier_returns_none_or_known_price(s):
  |                    ^^^
  |   File "C:\Users\admin\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 81, in test_tier_returns_none_or_known_price
    |     assert calculate_lift_access_tier(s) in (None, 35.0, 60.0, 85.0, 110.0)
    |            ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 24, in calculate_lift_access_tier
    |     return zone_rates[prefix]
    |            ~~~~~~~~~~^^^^^^^^
    | KeyError: ''
    | Failing test case: test_tier_returns_none_or_known_price(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\test_processor.py", line 81, in test_tier_returns_none_or_known_price
    |     assert calculate_lift_access_tier(s) in (None, 35.0, 60.0, 85.0, 110.0)
    |            ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\admin\L6_L7_08_ski_resort_pass\Lab 7\processor.py", line 17, in calculate_lift_access_tier
    |     prefix = zone_code.strip().upper()[:3]
    |              ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    | Failing test case: test_tier_returns_none_or_known_price(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
================================================ short test summary info =================================================
FAILED test_processor.py::test_parse_season_pass_validity_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_compute_trail_gradient_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_calculate_lift_access_tier_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_parse_returns_string - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
```